What Is Python?

Introduction

Python is a **high-level, general-purpose programming language** used to create software, automate tasks, process data, build websites, develop APIs, create desktop applications, perform scientific computing, and develop machine learning and artificial intelligence systems.

Python is designed with a focus on **readability, simplicity, and developer productivity**.

A simple Python program can look like this:

```
print("Hello, World!")
```

This program displays:

```
Hello, World!
```

Because Python syntax is relatively easy to read, it is widely used by beginners as well as professional software developers, data scientists, engineers, researchers, and automation developers.

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1. Definition of Python

Python is a **high-level, interpreted, dynamically typed, general-purpose programming language**.

Each part of this definition describes an important characteristic of Python.

Term	Meaning
High-level	Python provides abstractions that make programming easier for humans
Interpreted	Python programs are executed by a Python implementation such as CPython
Dynamically typed	Variable types are determined at runtime
General-purpose	Python can be used for many different types of applications
Programming language	Python provides syntax and rules for writing computer programs

A practical definition is:

> **Python is a readable, general-purpose programming language designed to make software development productive and maintainable.**

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2. Why Is Python Popular?

Python became popular because it combines relatively simple syntax with a large ecosystem of libraries and tools.

Some major reasons include:

- Simple and readable syntax
- Large standard library
- Large third-party ecosystem
- Cross-platform support
- Strong community
- Extensive documentation
- Rapid development
- Wide range of applications
- Strong support for data science
- Strong support for machine learning and AI
- Excellent automation capabilities
- Web development frameworks
- Database connectivity
- Scientific and mathematical libraries

Python allows developers to focus more on solving problems and less on complicated language syntax.

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3. Python as a High-Level Language

Python is considered a **high-level programming language**.

High-level languages provide abstractions that hide many low-level details of computer hardware.

For example:

```
numbers = [10, 20, 30, 40]

total = sum(numbers)

print(total)
```

Output:

```
100
```

The programmer does not need to manually manage CPU registers or memory addresses to calculate the sum.

Python provides higher-level abstractions such as:

- Variables
- Lists
- Dictionaries
- Functions
- Classes
- Modules
- Exceptions
- Iterators
- Generators

These abstractions make programs easier to write and understand.

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4. Python Is General-Purpose

Python is a **general-purpose programming language**.

This means Python is not designed for only one particular type of problem.

Python can be used for:

```
Web Development
        ↓
API Development
        ↓
Automation
        ↓
Data Analysis
        ↓
Machine Learning
        ↓
Artificial Intelligence
        ↓
Scientific Computing
        ↓
Desktop Applications
        ↓
Testing
        ↓
DevOps
```

For example, Python can be used to:

- Build a website
- Create a REST API
- Analyze financial data
- Automate repetitive tasks
- Process files
- Work with databases
- Train machine learning models
- Build command-line tools
- Perform scientific calculations
- Create test automation

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5. Python Syntax

Python syntax is designed to be readable.

For example:

```
name = "Sachin"

print(name)
```

Output:

```
Sachin
```

The code is relatively close to natural language compared with many lower-level programming languages.

Another example:

```
age = 20

if age >= 18:
    print("Adult")
```

Output:

```
Adult
```

Python uses **indentation** to define blocks of code.

The indentation in the example above is significant.

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6. Python Uses Indentation

Python uses indentation to organize blocks of code.

Example:

```
age = 25

if age >= 18:
    print("You are an adult.")
```

The statement inside the `if` block is indented.

Incorrect indentation can produce an error:

```
age = 25

if age >= 18:
print("You are an adult.")
```

Python will report an indentation-related error.

The standard style is to use **four spaces** for each indentation level.

Example:

```
if True:
    print("Level 1")

    if True:
        print("Level 2")
```

Indentation is an important part of Python syntax.

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7. Python Is Dynamically Typed

Python is a **dynamically typed language**.

You do not normally have to declare the type of a variable before assigning a value.

Example:

```
age = 25
name = "Sachin"
price = 99.99
is_active = True
```

Python determines the types of these values at runtime.

You can inspect a value's type using `type()`:

```
age = 25

print(type(age))
```

Output:

```
<class 'int'>
```

Another example:

```
name = "Python"

print(type(name))
```

Output:

```
<class 'str'>
```

Dynamic typing makes Python concise and flexible.

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8. Python Is Case-Sensitive

Python is **case-sensitive**.

This means uppercase and lowercase letters are treated differently.

For example:

```
name = "Python"

print(name)
```

works correctly.

But:

```
name = "Python"

print(Name)
```

produces an error because `name` and `Name` are different identifiers.

These are different:

```
name
Name
NAME
nAmE
```

Good Python code should use consistent naming conventions.

---

9. Python Is Cross-Platform

Python can run on many operating systems.

Common platforms include:

- Windows
- Linux
- macOS
- Other Unix-like systems

For example, the same basic Python program can generally be executed on Windows and Linux:

```
print("Hello from Python")
```

However, programs that interact directly with operating-system-specific features may require platform-specific code.

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10. Python and the Python Interpreter

Python source code is processed by a **Python implementation**.

A commonly used implementation is **CPython**.

For example, when you execute:

```
python hello.py
```

the Python implementation loads and processes the Python source code.

The exact execution process is more complex than simply saying that Python code is “interpreted.”

A simplified conceptual view is:

```
Python Source Code
        ↓
Python Implementation
        ↓
Bytecode / Execution Process
        ↓
Program Output
```

The details depend on the Python implementation.

---

11. Python Programs

A Python source file normally uses the `.py` extension.

Example:

```
hello.py
```

The file might contain:

```
print("Hello, Python!")
```

You can run it from a terminal with:

```
python hello.py
```

Depending on the operating system and Python installation, the command may also be:

```
python3 hello.py
```

On Windows, `py` may also be available:

```
py hello.py
```

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12. Python Interactive Mode

Python can also be used interactively.

Running:

```
python
```

may open the Python interactive interpreter.

You may see something similar to:

```
>>>
```

You can then enter Python code directly:

```
>>> 10 + 20
30
```

Another example:

```
>>> print("Hello")
Hello
```

Interactive mode is useful for:

- Learning Python
- Testing small pieces of code
- Experimenting with expressions
- Checking library behavior
- Debugging simple problems

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13. Python Standard Library

Python includes a large **standard library**.

The standard library provides modules for many common programming tasks.

Examples include:

```
import math

print(math.sqrt(25))
```

Output:

```
5.0
```

Another example:

```
import datetime

print(datetime.datetime.now())
```

The standard library includes functionality related to:

- Mathematics
- Dates and times
- Files
- Operating systems
- Networking
- JSON
- Regular expressions
- Databases
- Testing
- Logging
- Concurrency
- Data formats

This reduces the need to implement common functionality from scratch.

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14. Python Third-Party Libraries

Python also has a huge ecosystem of third-party packages.

Examples include:

Library	Common Use
NumPy	Numerical computing
Pandas	Data analysis
Matplotlib	Data visualization
Plotly	Interactive visualization
SciPy	Scientific computing
scikit-learn	Machine learning
TensorFlow	Machine learning and deep learning
PyTorch	Machine learning and deep learning
OpenCV	Computer vision
Keras	Deep learning
Requests	HTTP requests
FastAPI	API development
Django	Web development
Flask	Web development

These packages extend Python into many specialized fields.

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15. Python Package Management

Python packages can commonly be installed using **pip**.

For example:

```
python -m pip install requests
```

After installation, a package can be imported into a Python program:

```
import requests
```

Package management is an important part of real-world Python development.

It will be studied later in:

```
06_tools_libraries/
└── 25_pip_package_management/
```

16. Python and Web Development

Python is widely used for web development.

Popular Python web frameworks include:

- Django
- Flask
- FastAPI

For example, FastAPI can be used to create APIs.

A simple example:

```
from fastapi import FastAPI

app = FastAPI()

@app.get("/")
def home():
    return {"message": "Hello, Python"}
```

Python can therefore be used to build:

- Websites
- REST APIs
- Backend services
- Microservices
- Web applications

17. Python and Data Science

Python is widely used for data analysis and scientific computing.

For example:

```
import pandas as pd

data = {
    "name": ["Alice", "Bob", "Charlie"],
    "score": [85, 90, 78]
}

df = pd.DataFrame(data)

print(df)
```

Libraries such as:

- NumPy
- Pandas
- SciPy
- Matplotlib
- Seaborn
- Plotly
- Statsmodels

provide tools for working with data.

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18. Python and Machine Learning

Python is heavily used in machine learning.

Popular libraries include:

- scikit-learn
- TensorFlow
- Keras
- PyTorch

A simple machine-learning workflow may involve:

```
Collect Data
    ↓
Clean Data
    ↓
Explore Data
    ↓
Prepare Features
    ↓
Train Model
    ↓
Evaluate Model
    ↓
Make Predictions
```

Python's libraries make these workflows easier to implement.

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19. Python and Automation

Python is excellent for automation.

For example, Python can automate:

- File operations
- Folder organization
- Data processing
- Report generation
- Web tasks
- Testing
- System administration
- Scheduled jobs
- Data collection

Example:

```
from pathlib import Path

for file in Path(".").iterdir():
    print(file)
```

This program lists files and directories in the current directory.

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20. Python and Database Programming

Python can communicate with databases.

Common database technologies used with Python include:

- SQLite
- PostgreSQL
- MySQL
- Microsoft SQL Server
- MongoDB

For example, Python includes built-in support for SQLite:

```
import sqlite3

connection = sqlite3.connect("example.db")

print("Database connected")

connection.close()
```

Database programming becomes especially useful when building real applications.

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21. Python and Scientific Computing

Python is widely used in scientific and engineering applications.

It can be used for:

- Mathematics
- Statistics
- Simulations
- Numerical analysis
- Data processing
- Scientific visualization
- Research
- Engineering calculations

Libraries such as NumPy and SciPy provide powerful numerical and scientific functionality.

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22. Python and Artificial Intelligence

Python is one of the major programming languages used in modern AI development.

Python can be used for:

- Machine learning
- Deep learning
- Natural language processing
- Computer vision
- Data preparation
- Model training
- Model evaluation
- AI experimentation

Libraries such as PyTorch, TensorFlow, Keras, and scikit-learn provide tools for these areas.

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23. Python Community

Python has a large global developer community.

The community contributes:

- Libraries
- Frameworks
- Tutorials
- Documentation
- Open-source projects
- Tools
- Educational resources
- Conferences

A large community makes it easier to find learning resources and solutions to common programming problems.

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24. Advantages of Python

Important advantages include:

24.1 Readable Syntax

Python code is generally easy to read.

```
total = price * quantity
```

24.2 Productivity

Python allows developers to write useful programs with relatively little code.

24.3 Large Ecosystem

Thousands of packages extend Python’s capabilities.

24.4 Versatility

Python can be used across many domains.

24.5 Cross-Platform

Python is available on major operating systems.

24.6 Strong Community

There is extensive community support and educational material.

24.7 Rapid Development

Python is well suited to prototyping and iterative development.

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25. Limitations of Python

Python is powerful, but it is not perfect.

Some limitations include:

25.1 Performance

For some CPU-intensive workloads, Python can be slower than compiled languages such as C or C++.

25.2 Memory Usage

Python programs can use more memory than equivalent programs written in some lower-level languages.

25.3 Mobile Development

Python is not generally the primary choice for native mobile application development.

25.4 Browser Frontend Development

Python is not normally used as the primary language for browser-side frontend programming.

JavaScript and related technologies dominate that area.

25.5 Runtime Type Errors

Because Python is dynamically typed, some type-related problems may only become apparent when the program runs.

Modern Python provides type hints and tools that can help catch many such problems earlier.

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26. Where Python Is Used

Python is used in many industries and technical areas.

```
Python
├── Web Development
├── API Development
├── Data Science
├── Machine Learning
├── Artificial Intelligence
├── Automation
├── Scientific Computing
├── Finance
├── Cybersecurity
├── Testing
├── DevOps
├── Education
├── Research
├── Computer Vision
├── Natural Language Processing
└── Data Engineering
```

Python is therefore not limited to one career or technology area.

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27. Simple Python Example

Here is a small Python program:

```python
name = "Python"
version = 3

print("Language:", name)
print("Major Version:", version)
```

Output:

```
Language: Python
Major Version: 3
```

This simple program demonstrates:

- Variables
- Strings
- Integers
- Assignment
- `print()`
- Program execution

These concepts will be studied in greater detail in later lessons.

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28. Python File Extension

Python source files normally use:

```
.py
```

Examples:

```
hello.py
calculator.py
main.py
app.py
server.py
```

Jupyter Notebook files commonly use:

```
.ipynb
```

Example:

```
introduction_to_python.ipynb
```

---

29. Python’s Role in Your Learning Path

In this curriculum, Python will progress from basic concepts to professional development.

The progression is approximately:

```
Python Fundamentals
        ↓
Data Structures
        ↓
Control Flow
        ↓
Functions and Modules
        ↓
Advanced Concepts
        ↓
Tools and Libraries
        ↓
Python Libraries
        ↓
Specialized Topics
        ↓
Applications
        ↓
Professional Development
        ↓
Projects and Interview Preparation
```

The first stage focuses on understanding the language itself.

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30. What You Should Know After This Lesson

After studying this lesson, you should be able to explain:

1. What Python is
2. Why Python is popular
3. What a high-level programming language is
4. What a general-purpose programming language is
5. What dynamic typing means
6. Why Python uses indentation
7. What a Python source file is
8. What the `.py` extension means
9. What the Python interpreter does
10. What Python's interactive mode is
11. What the Python standard library is
12. What third-party packages are
13. Where Python is commonly used
14. Some advantages of Python
15. Some limitations of Python

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31. Key Terms

Term	Definition
Python	A general-purpose programming language
Programming Language	A language used to give instructions to computers
High-Level Language	A language that provides abstractions closer to human concepts
Interpreter	Software that executes or otherwise processes program code
CPython	The reference implementation of Python, primarily written in C
Dynamic Typing	Types are associated with runtime values rather than requiring variable declarations
Syntax	Rules that define how Python code is written
Variable	A name used to refer to a value
Module	A Python file containing code that can be imported
Package	A distribution or organized collection of Python modules/packages
Library	Reusable code that provides functionality to applications
Framework	A structured foundation for building applications
REPL	Read-Eval-Print Loop for interactive programming
Bytecode	An intermediate representation used by CPython's execution process

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32. Summary

Python is a **high-level, general-purpose programming language** known for its readable syntax and broad ecosystem.

It can be used for:

- Software development
- Web development
- API development
- Automation
- Data analysis
- Scientific computing
- Machine learning
- Artificial intelligence
- Computer vision
- Testing
- Database programming

Python supports both beginners and experienced developers because it combines relatively simple syntax with powerful libraries, frameworks, and development tools.

The fundamental idea to remember is:

> **Python is a general-purpose programming language designed to make it easier to write, understand, and maintain software.**

The next lesson examines the **history of Python** and how the language evolved from its early development into the modern Python ecosystem.

Quick Review

Q: What is Python?

Python is a high-level, general-purpose programming language used for a wide range of software-development tasks.

Q: Is Python only used for web development?

No. Python is used for web development, automation, data science, machine learning, scientific computing, testing, APIs, and many other areas.

Q: What file extension is normally used for Python source code?

`.py`

Q: What is CPython?

CPython is the reference implementation of Python and is primarily implemented in the C programming language.

Q: Is Python case-sensitive?

Yes.

Q: Does Python use indentation?

Yes. Indentation is significant in Python syntax.

Q: Is Python dynamically typed?

Yes. Python determines and tracks types at runtime rather than requiring traditional static variable declarations.

Q: Can Python work with databases?

Yes. Python supports databases such as SQLite, PostgreSQL, MySQL, and others through built-in modules and third-party packages.

Q: Can Python be used for AI and machine learning?

Yes. Python has a large ecosystem for machine learning and AI, including libraries such as scikit-learn, PyTorch, TensorFlow, and Keras.